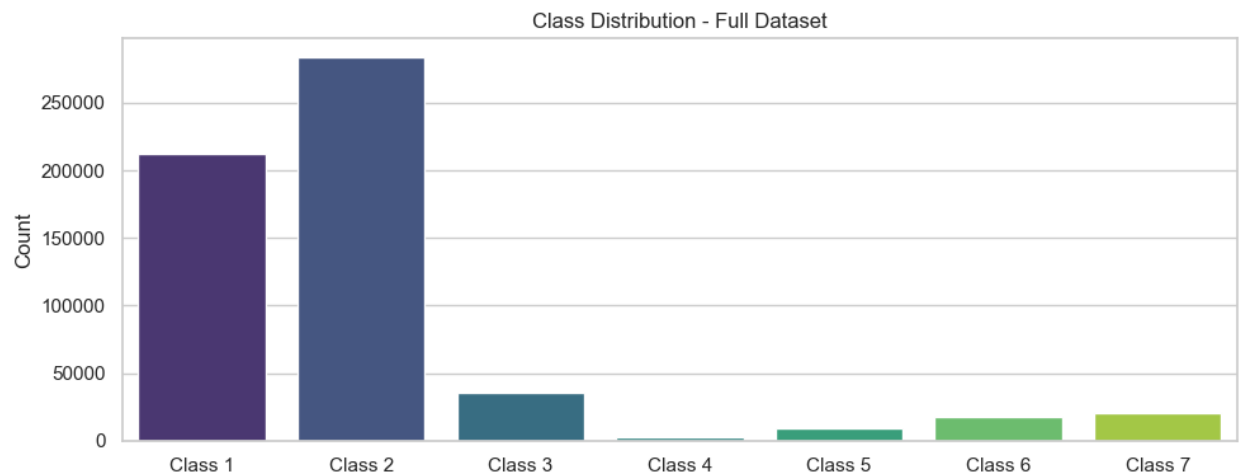


Project 2 Report: Forest Cover Type Classification with DNNs

Baseline Results

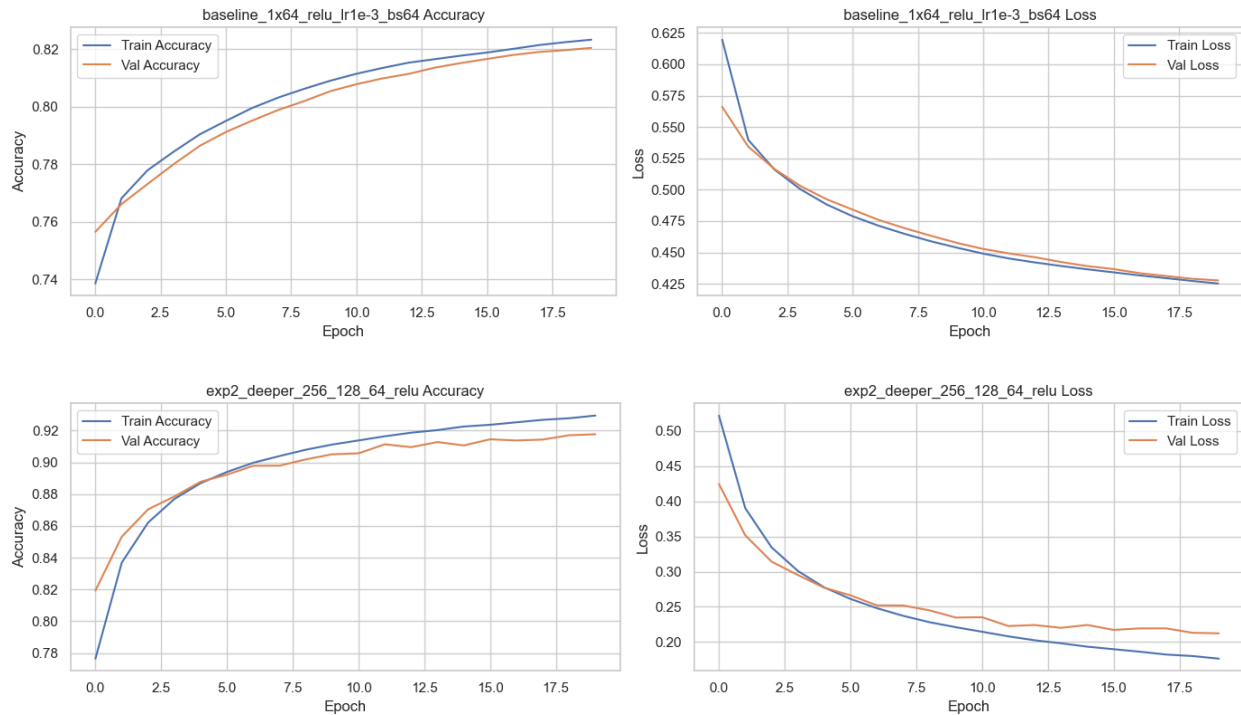
I used a baseline dense neural network with one hidden layer of 64 neurons, ReLU activation, no dropout, learning rate 0.001, batch size 64, and 20 epochs. This model gave me a useful starting point for comparison because it was simple and easy to interpret. The baseline reached a training accuracy of 0.823, validation accuracy of 0.820. From the learning curves, the model showed stable convergence with room to improve. Validation and training accuracy were close, suggesting the model was not overfitting but also had limited capacity. That helped me decide to try deeper architecture and other changes next.



Effects of Hyperparameter Changes

I changed one main factor at a time to keep comparisons fair. Increasing depth from one hidden layer to two (128–64) gave a clear jump, the best validation accuracy went from 0.820 to 0.889. Going to three layers (256–128–64) improved further to 0.918, the best of all experiments, with train accuracy about 0.929 and a small gap between train and val, so the deeper network generalized well. Switching to ELU (128–64) gave validation 0.877, slightly below the same-sized ReLU model, so ReLU was at least as good for this setup. Adding dropout 0.3 (128–64) reduced overfitting, train 0.805, val best 0.838, but also lowered validation accuracy compared to no dropout (0.889), so the regularization was a bit strong for 20 epochs. L2 (1e-4) on 128–64 gave a val accuracy of 0.880, a small gain over baseline and similar in spirit to dropout, slightly constraining the model. Lower learning rate (5e-4)

and larger batch (128) gave a validation accuracy of 0.86 so slower updates and bigger batches did not help here and likely slowed adaptation.

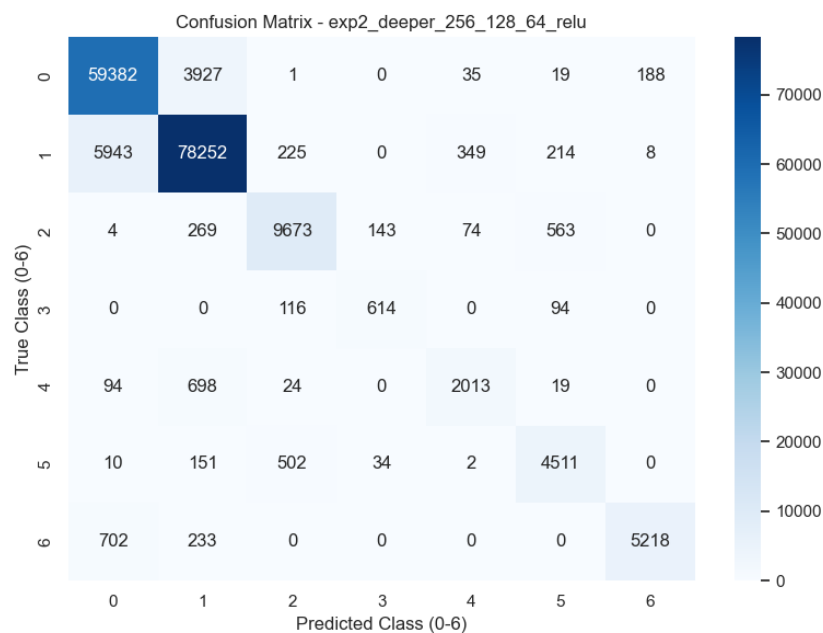


experim ent	val_accuracy_last	val_accuracy _best	train_accurac y_last	val_loss_ last	
2	exp2_deeper_256_128_64_relu	0.917644	0.917644	0.929418	0.212181
1	exp1_deeper_128_64_relu	0.889073	0.889073	0.892352	0.274305
5	exp5_l2_1e-4	0.880185	0.880185	0.883390	0.344785
3	exp3_activation_elu	0.877431	0.877431	0.879536	0.296964
6	exp6_lr5e-4_bs128	0.865543	0.865543	0.871744	0.328553
4	exp4_dropout_0p3	0.838140	0.838251	0.805072	0.384273

experiment	val_accuracy_last	val_accuracy_best	train_accuracy_last	val_loss_last	
0	baseline_1x64_relu_lr1e-3_bs64	0.820413	0.820413	0.823254	0.427687

Best Model

The best-performing model was exp2_deeper_256_128_64_ with yjr configuration three hidden layers (256, 128, 64), ReLU activation, no dropout, no L2, learning rate 0.001, batch size 64, trained for 20 epochs. I selected this model because it had the highest validation accuracy (0.918) and maintained strong test accuracy (0.916). The curves looked more stable than the baseline and other experiments, with a small train/validation gap. Its final test accuracy was 0.916, and class level metrics were more balanced than the baseline.



Reflections

What worked:

Deeper architecture (256–128–64) gave the biggest gain without hurting generalization. Going from 1 to 2 to 3 hidden layers improved validation consistently. ReLU was at least as good as ELU in the 128–64 comparison. The best model did not need dropout or L2 for this dataset and epoch budget.

Christian Basso

What did not help:

Dropout 0.3 was too strong and lowered validation accuracy. Lower learning rate and larger batch size reduced performance. L2 gave a small improvement over baseline but did not beat the deeper ReLU model without regularization.

Next steps:

I would add early stopping and a learning-rate scheduler, and possibly class weights or oversampling for classes 3 and 4 to improve their recall. Trying one or two more deeper/wider variants and more epochs with early stopping could also help.